BeatAhead Model Card: Phase 5 Deployment Candidate (XGBoost Matrix A)

Model Name: `beatahead_phase5_model`
Model Version: `1.0.0`
Feature Schema Version: `1.0.0`
Document Date: 2026-09-22
Framework: XGBoost (`v3.4.1`) & Scikit-Learn (`v1.9.1`)
Deployment Artifact: `models/beatahead_phase5_model.joblib`
Status: FROZEN RESEARCH PROTOTYPE — NOT CLINICALLY APPROVED FOR DIAGNOSTIC USE

1. Regulatory & Safety Disclaimer

> [!CAUTION] **INVESTIGATIONAL RESEARCH PROTOTYPE ONLY**:
> The BeatAhead physiological machine learning model is an experimental research prototype developed for retrospective algorithmic benchmarking on open-access surgical biosignals (VitalDB). It has **NOT** been evaluated in prospective clinical trials, has **NOT** received regulatory clearance (e.g., FDA 510(k), De Novo, or CE mark), and must **NEVER** be used as a primary diagnostic tool, standalone clinical monitor, or automated therapeutic trigger in patient care.

2. Intended Use & Clinical Scope

- **Primary Clinical Task**: Early warning of impending intraoperative myocardial ischemia during surgical procedures under general anesthesia.
- **Target Population**: Adult patients undergoing major non-cardiac or cardiovascular surgery with continuous multi-parameter monitoring (ECG Lead II, finger photoplethysmogram, pulse oximetry).
- **Operational Lead Time**: Generates predictions with an enforced **5-minute advance lead time** ($T_{\text{obs}} = 300\text{ s}$, $T_{\text{gap}} = 300\text{ s}$, $T_{\text{target}} = 300\text{ s}$).
- **Intended Clinical Context**: Secondary decision-support alert designed to notify anesthesia teams of subtle, insidious ST segment micro-drifting before overt clinical thresholds ($\le -1.0\text{ mm}$) are breached.
- **Out of Scope**:
- Out-of-hospital or ambulatory cardiac monitoring (e.g. smartwatches or patch ECGs).
- Diagnostic localization of regional wall motion abnormalities (e.g., distinguishing LAD vs RCA occlusions).
- Detection of non-ischemic arrhythmias or hemodynamic shock without repolarization changes.

3. Experimental Target Definition

The ground truth prediction target is defined as: $$\text{Target } y = 1 \iff \max\text{consecutive_duration} (\text{ST_II} \le -1.0\text{ mm}) \ge 60\text{ seconds within } T_{\text{target}}$$

Temporal Architecture Constraints:

- **Observation Window (T_{obs})**: $[t, t + 300\text{ s})$ — raw signal window from which all 26 features are exclusively extracted.
- **Prediction Lead-Time Gap (T_{gap})**: $[t + 300\text{ s}, t + 600\text{ s})$ — mandatory blank gap with zero feature extraction, ensuring true early warning.
- **Target Horizon (T_{target})**: $[t + 600\text{ s}, t + 900\text{ s})$ — interval where ground truth sustained ischemia is evaluated.
- **Sliding Stride**: 60 seconds between overlapping evaluation windows.
- **Exclusion of Contaminated Windows**: Windows with active sustained ischemia ($\ge 60\text{ s}$) inside T_{obs} or T_{gap}, or within 15 minutes post-resolution, are labeled $y = -1$ and strictly excluded.

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4. Training Dataset & Cohort Composition

The model was developed on the official **100-Case VitalDB Surgical Dataset**: - **Cohort A (High-Risk Cardiovascular Enriched, $N=50$)**: Elderly patients ($\ge 65\text{ yrs}$) with documented hypertension (`preop_htn = 1`), diabetes (`preop_dm = 1`), or vascular/CAD diagnosis. - **Cohort B (Representative Multi-Specialty Surgical, $N=50$)**: Uniformly sampled across general, thoracic, urology, and gynecology surgical departments, completely blind to ST outcomes. - **Replacement Case 33**: Pre-audited Case 33 (Hernia repair, 1.75h) replaced defective candidate Case 5326 (which lacked `SNUADC/PLETH` upstream). - **Two-Tier Subject-Level Partition**: - **Development Set ($N = 80$ Cases; 40 A, 40 B)**: 8,571 usable windows, 37 positive ($y=1$), 8,534 clean negative ($y=0$); 6 positive patients (`[137, 864, 2172, 4236, 4390, 4568]`). - **Locked Test Set ($N = 20$ Cases; 10 A, 10 B)**: 2,770 usable windows, 11 positive ($y=1$), 2,759 clean negative ($y=0$); 2 positive patients (`[2085, 3280]`). - **Zero Patient Overlap**: $\text{Dev} \cap \text{Test} = \emptyset$.

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5. Input Feature Schema (Matrix A, 26 Features)

All features are extracted strictly from T_{obs} and validated against physiological bounds:

#	Feature Name	Modality	Unit	Valid Bounds	Description
1	`ecg_hr_mean`	ECG II	bpm	$[20.0, 300.0]$	Mean heart rate over 5-minute window
2	`ecg_hr_std`	ECG II	bpm	$[0.0, 150.0]$	Heart rate standard deviation (autonomic tone)
3	`ecg_rr_sdnn`	ECG II	ms	$[0.0, 1000.0]$	SDNN: Standard deviation of normal RR intervals
4	`ecg_rr_rmssd`	ECG II	ms	$[0.0, 1000.0]$	RMSSD: Successive RR interval difference root mean square
5	`ecg_pnn50`	ECG II	%	$[0.0, 100.0]$	Percentage of adjacent RR intervals differing by $>50\text{ ms}$
6	`ecg_r_amp_mv`	ECG II	mV	$[-2.0, 15.0]$	Median R-wave amplitude above isoelectric line
7	`ecg_qrs_width_ms`	ECG II	ms	$[30.0, 300.0]$	QRS complex duration estimate
8	`ecg_sqi`	ECG II	ratio	$[0.0, 1.0]$	Electrocardiogram signal quality index
9	`pat_median_ms`	ECG+PPG	ms	$[50.0, 600.0]$	Median Pulse Arrival Time (QRS peak to PPG foot delay)
10	`pat_iqr_ms`	ECG+PPG	ms	$[0.0, 500.0]$	Interquartile range of PAT (vascular transit stability)
11	`pat_valid_fraction`	ECG+PPG	ratio	$[0.0, 1.0]$	Fraction of cycles with plausible PAT (100–450 ms)
12	`pat_valid`	ECG+PPG	binary	$[0, 1]$	Binary validity flag ($\ge 30\%$ plausible PAT beats)
13	`ppg_pulse_amp`	PPG	a.u.	$[0.0, 50000.0]$	Median peak-to-trough plethysmogram pulse amplitude
14	`ppg_perfusion_index`	PPG	%	$[0.0, 10000.0]$	Pulsatile AC / non-pulsatile DC perfusion index
15	`ppg_crest_time_ms`	PPG	ms	$[20.0, 500.0]$	Foot to systolic peak crest time (arterial stiffness)
16	`ppg_sqi`	PPG	ratio	$[0.0, 1.0]$	Photoplethysmogram signal quality index
17	`spo2_mean`	SpO2	%	$[40.0, 100.0]$	Mean pulse oximeter oxygen saturation
18	`spo2_min`	SpO2	%	$[30.0, 100.0]$	Nadir oxygen saturation
19	`spo2_std`	SpO2	%	$[0.0, 40.0]$	Oxygen saturation standard deviation
20	`spo2_desat_count`	SpO2	s	$[0, 300]$	Duration of arterial desaturation ($\text{SpO2} \le 90\%$)
21	`st_obs_mean`	ST II	mm	$[-15.0, 15.0]$	Mean ST segment deviation in Lead II
22	`st_obs_median`	ST II	mm	$[-15.0, 15.0]$	Median ST segment deviation in Lead II

#	Feature Name	Modality	Unit	Valid Bounds	Description
23	`st_obs_min`	ST II	mm	$[-15.0, 15.0]$	Minimum ST segment deviation in Lead II
24	`st_obs_std`	ST II	mm	$[0.0, 15.0]$	ST segment standard deviation
25	`st_delta_baseline`	ST II	mm	$[-15.0, 15.0]$	ST shift relative to initial 10-minute surgical baseline
26	`st_slope_mm_min`	ST II	mm/min	$[-20.0, 20.0]$	ST regression slope across 5-minute window

6. Model Architecture & Hyperparameters

- **Algorithm**: `XGBClassifier` (Extreme Gradient Boosting Trees)
- **Objective Function**: `binary:logistic` (Logloss)
- **Class Imbalance Handling**: `scale_pos_weight = 230.65` (matching the empirical $8,534 / 37$ negative-to-positive ratio in development)
- **Tree Parameters**:
- `n_estimators`: 150
- `max_depth`: 4
- `learning_rate`: 0.03
- `subsample`: 0.8
- `colsample_bytree`: 0.8
- `random_state`: 42
- **Preprocessing Pipeline**: `SimpleImputer(strategy='median')` fit strictly on the 80 development cases.

7. Decision Threshold & Calibration Procedure

- **Model Probability (P_{model})**: Raw continuous output in $[0, 1]$ representing predicted likelihood of sustained ischemia within T_{target}.
- **Frozen Decision Threshold (T_{decision})**: **`0.156742`**.
- **Calibration Method**: Calibrated on pooled out-of-fold predictions from the 80 development cases to maximize F_1-score ($\text{argmax}(F_1)$).
- **Binary Decision Logic**: $$\text{alert} = \begin{cases} 1 \text{ ("ALERT")}, & \text{if } P_{\text{model}} \ge 0.156742 \ 0 \text{ ("NORMAL")}, & \text{if } P_{\text{model}} < 0.156742 \end{cases}$$

8. Frozen Benchmark Performance Metrics

Development 5-Fold Cross-Validation (80 Cases, OOF Pooled):

- **PR-AUC (Average Precision)**: **`0.3247`** (vs 0.0043 natural prevalence = **~75x above chance**)
- **Mean Fold PR-AUC**: **0.4812 ± 0.3834**

- **AUROC**: `0.8896`
- **Brier Calibration Loss**: `0.0045`

Locked Benchmark Test Set (20 Cases, Sealed):

- **Test Window Specificity**: `99.71%`
- **False Alarm Rate**: `0.1733 alarms / surgical hour` (1 false alarm every 5.77 hours)
- **Test Window Sensitivity**: `0.0%` (on abrupt-onset test cases)
- **Test PR-AUC**: `0.0043` | **Test AUROC**: `0.4967` | **Test Brier**: `0.0056`

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9. Limitations, Failure Modes & Observed Phenotypes

1. **Observed Phenotype Sensitivity (Insidious vs. Abrupt)**: - **Insidious Micro-Drifts**: The model accurately detects progressive subendocardial ischemia where pre-event ST drifts downward ($\text{mean ST} = -0.53\text{ mm}$ during T_{obs}). - **Abrupt Onset Collapses**: When ischemia occurs instantaneously (e.g. sudden cross-clamping or acute vasospasm), the ST segment is completely normal ($+0.27\text{ mm}$) 5 minutes prior. The model refuses to sound false alarms on normal physiology, resulting in zero early warning. A real-time deterministic reflex ($\text{ST} \le -1.0\text{ mm}$) must operate in tandem.
2. **Failure of Pure Non-ST Features (Matrix B Ablation)**: Ablation benchmarks proved that non-ST vascular and autonomic features alone (PAT, HRV, pulse morphology) achieve an AUROC of only 0.53. BeatAhead requires electrophysiological repolarization features (ST segment) to achieve clinical utility.
3. **Single-Lead II Monitoring**: Ischemia limited to the circumflex or left anterior descending distribution may express exclusively in precordial leads (e.g. V5). Events without Lead II manifestations will not be detected.
4. **Motion & Surgical Electrocautery**: Electrocautery creates high-frequency bursts that degrade `ecg_sqi` and `ppg_sqi`. Severe artifact windows are rejected by SQI filters.